

## Multiple Choice (10 marks)

1. The half-life of radon is 3.80 days. After how many days will only one-sixteenth of a radon sample remain?
  - (a) 26.6 days
  - (b) 19.0 days
  - (c) 3.80 days
  - (d) 5.7 days
  - (e) 22.8 days
2. What is the intensity of the electric field 10 cm from a negative point charge of 500 stat-coulombs in air?
  - (a) 5 dyne/stat-coul
  - (b) 8 dyne/stat-coul
  - (c) 0.5 dyne/stat-coul
  - (d) 100 dyne/stat-coul
  - (e) 10 dyne/stat-coul
3. Calculate the mean ionic activity of a 0.0350 mNa<sub>3</sub>PO<sub>4</sub> solution for which the mean activity coefficient is 0.685 .
  - (a) 0.0612
  - (b) 0.0426
  - (c) 0.0753
  - (d) 0.0905
  - (e) 0.0294
4. determine the ratio of the radius of a uranium-238 nucleus to the radius of a helium-4 nucleus.
  - (a) 4.8
  - (b) 7.1
  - (c) 5.5
  - (d) 1.8
  - (e) 4.2
5. The Pleiades is an open star cluster that plays a role in many ancient stories and is well-known for containing ... bright stars.

- (a) 5
- (b) 7
- (c) 9
- (d) 12

## True / False (10 marks)

*Answer True or False*

6. True or False: The correct answer to 'What equal positive charges would have to be placed on Earth and on the Moon to neutralize their gravitational attraction?' is ' $3.3 \times 10^{13} \text{C}$ '.
7. True or False: Meteorites always exhibit a high density due to their unique composition and formation process.
8. True or False: The potential energy of a compressed spring and a charged object depends solely on the distance they travel.
9. True or False: A completely submerged object displaces a volume of fluid equal to the pressure exerted by the fluid on its surface.
10. True or False: A voltage of 10 volts is sufficient to send a current of 2 amperes through a circuit with a resistance of 3 ohms.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Calculate the net electric flux through a Gaussian cube with an edge length of 55 cm, centered around a particle with a charge of  $1.8 \mu\text{C}$ . Discuss the principles involved.
12. A child slides a block of mass 2 kg at an initial speed of 4 m/s towards a spring with a spring constant of 6 N/m. Analyze the energy transformations involved and determine the maximum compression of the spring, providing a detailed explanation of your reasoning and calculations.

*End of Paper*